

Data Modelling

In today's data-driven world, organisations collect and manage vast amounts of data. Structuring and organising this data efficiently is crucial for **effective storage, retrieval, and analysis**. Data modelling serves as the blueprint for this process, guiding how data is stored and accessed, and ensuring it is organised logically.

Imagine working on a global initiative aiming to improve healthcare access and quality worldwide, aligning with the United Nations Sustainable Development Goal 3: *Good Health and Well-being*. The data collected from various countries, hospitals, and patient records needs to be structured effectively to make informed decisions. Throughout this train, we'll use the healthcare sector as an example to illustrate key concepts, enhancing your understanding of different data modelling techniques and architectures.

Learning Objectives

By the end of this train, you should be able to:

1. Understand the significance of data modelling in enhancing data storage, retrieval, and analysis efficiency.
2. Explore the principles of Third Normal Form (3NF) for reducing data redundancy and dependency.
3. Understand the Data Vault methodology for enabling scalable and flexible data warehouses.
4. Gain insights into the star schema for simplifying complex data queries in a dimensional model.

Why is Data Modelling Important?

Data modelling helps reduce redundancy and ensures consistency, leading to efficient data storage and faster retrieval. This efficiency is vital for real-time analytics and decision-making.

A well-designed data model simplifies complex queries and analytical processes, making it easier to extract meaningful insights from large datasets.

Example: World Health Organization (WHO)

Massive organisations like the WHO rely on structured data models to quickly access and analyse health data from different countries, facilitating rapid and effective responses to health crises.

Exploring Data Modelling Techniques

Let's delve into various data modelling techniques, understanding their principles, strengths, and applications.

Third Normal Form (3NF)

Third Normal Form is a database normalisation technique used to reduce data redundancy and improve data integrity in relational databases. It ensures that every non-primary key attribute is directly dependent on the primary key and not on other non-primary attributes.

Key principles

- **Eliminate redundant data:** Ensure that each piece of information is stored only once in the database to avoid duplication. This reduces storage costs and minimises the risk of inconsistencies arising from multiple copies of the same data.
- **Ensure logical dependencies:** Ensure that all attributes in a table depend solely on the primary key and not on any other non-primary attributes. This simplifies data maintenance and enhances data integrity by preventing anomalies.

Benefits of 3NF

- **Data integrity:** Reduces anomalies during data insertion, updating, and deletion, ensuring that the data remains accurate and reliable. This is crucial for making informed decisions based on trustworthy information.
- **Efficient storage:** Saves disk space by eliminating duplicate data entries, which optimises the use of storage resources. Efficient storage also contributes to improved system performance and scalability.
- **Simplified Maintenance:** Makes it easier to manage and update data structures because changes need to be made in only one place. This reduces the likelihood of errors and streamlines database administration tasks.

Example: Patient records management

In a hospital database, applying Third Normal Form (3NF) ensures that patient information and treatment data are stored without redundancy, maintaining data integrity and consistency.

Patient Table

This table stores unique information about each patient.

PatientID (PK)	FirstName	LastName	DateOfBirth	Gender	ContactNumber
P001	Loyiso	Cekiso	15/05/1980	Male	0712345678
P002	Qhamasande	Matya	22/08/1990	Female	0723456789

- **Primary Key (PK):** PatientID
- **Attributes:** Depend solely on PatientID.

Treatment Table

This table records treatments administered to patients.

TreatmentID (PK)	PatientID (FK)	TreatmentDate	TreatmentType
T001	P001	10/01/2023	X-Ray
T002	P002	12/01/2023	Blood Test
T003	P001	15/01/2023	MRI Scan

- **Primary Key (PK):** TreatmentID
- **Foreign Key (FK):** PatientID references Patient.PatientID
- **Attributes:** Depend solely on the primary key.

Why are these tables in Third Normal Form?

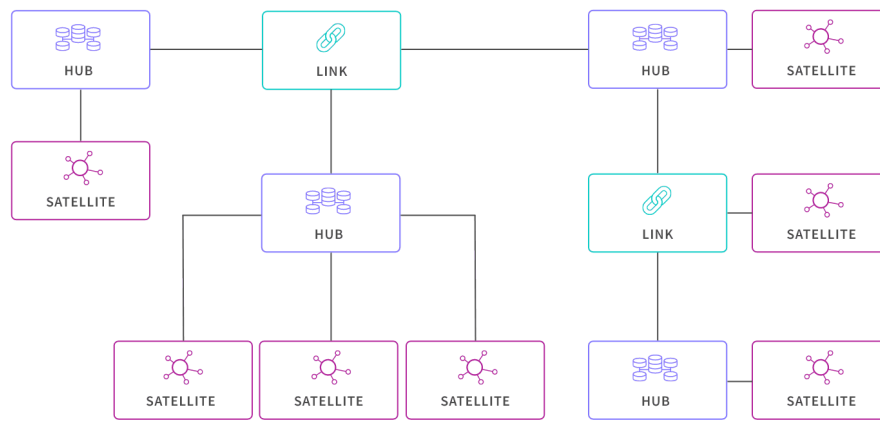
- In the Patient Table, all non-key attributes (FirstName, LastName, DateOfBirth, Gender, ContactNumber) depend directly on the primary key (PatientID). There is no transitive dependency (no non-key column depends on another non-key column).
- In the Treatment Table, the TreatmentDate and TreatmentType depend on the primary key (TreatmentID), and the PatientID is a foreign key. There is no transitive dependency here either.

Data Vault Modelling

Data Vault is a methodology for designing data warehouses that can handle large volumes of historical data from multiple sources. It is designed for scalability and flexibility, accommodating changes in data structures over time.

Data vault core components

- **Hubs:** These tables store unique business keys, such as patient IDs or product codes, which are critical identifiers across the organisation. Hubs provide a central point of reference for linking related data without duplication.
- **Links:** Links represent relationships between hubs, capturing associations such as patient treatments by doctors or patient admissions to hospitals.
- **Satellites:** Satellites contain descriptive data and timestamps, storing the attributes and context related to hubs and links. They allow for tracking historical changes and provide the ability to audit data over time.



Source: <https://www.qlik.com/us/data-warehouse/data-vault>

Example

A national health agency may use a Data Vault to consolidate patient data from various hospitals, integrating disparate data sources while preserving historical changes.

Benefits of data vault

- **Scalability:** Data Vault can handle growing amounts of data efficiently, making it suitable for large-scale data warehousing needs. Its architecture supports parallel loading and processing, accommodating increased data volumes without performance degradation.
- **Flexibility:** It easily adapts to new data sources and changes in existing ones, allowing organisations to evolve their data infrastructure without extensive rework. This adaptability reduces the time and cost associated with integrating new data.
- **Auditability:** Data Vault maintains a complete history of data changes, providing a transparent audit trail. This is essential for compliance with regulations and for understanding how data has evolved over time.

Reflect:

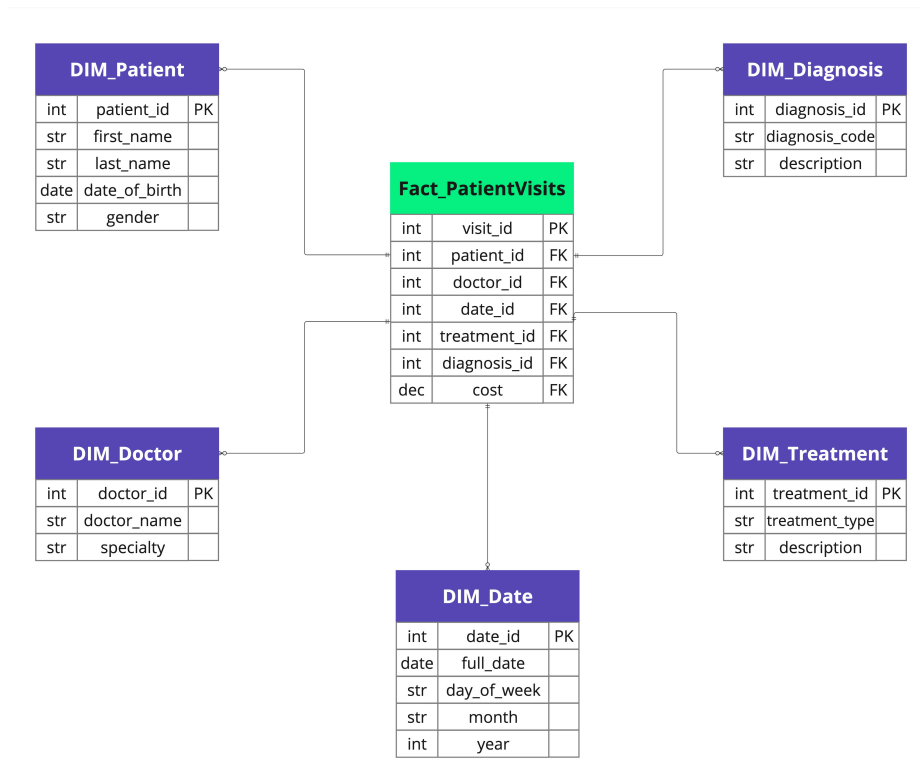
What challenges are faced in integrating data from different sources and how might a Data Vault help in maintaining data quality and consistency?

Star Schema

The star schema is a data modelling technique used in data warehousing for optimising query performance. It features a central fact table connected to multiple dimension tables, resembling a star shape.

Star schema structure

- **Fact table:** Contains quantitative data about events. For example, it could record metrics like the number of patient admissions, treatments provided, or vaccination counts.
- **Dimension tables:** Hold descriptive attributes that provide context to the facts. Examples include patient information (age, gender) or healthcare facilities (location, capacity). These tables allow analysts to explore the data from different perspectives.



Benefits of star schema

- **Simplified queries:** Facilitates straightforward SQL queries for reporting and analysis.
- **Performance:** Optimised for read-intensive operations, leading to faster query execution and data retrieval.
- **User-friendly:** Easier for data analysts to understand and navigate.

When to Select which Technique?

Data modelling is essential for structuring and organising data to ensure efficient storage, retrieval, and analysis. Different techniques offer unique strengths tailored to different needs.

Before flipping the cards, take a moment to consider which technique you would choose based on the following requirements:

Scalability and historical tracking

Implement Data Vault for integrating multiple data sources over time.

Data integrity and consistency

Use 3NF for operational systems where accuracy is paramount.

Performance and simplicity

Choose star schema for analytical applications requiring fast query responses.

Conclusion

Data modelling is a foundational practice for organising and structuring data to enable efficient storage, retrieval, and analysis, ultimately supporting timely and informed decision-making. By understanding and applying techniques such as Third Normal Form, Data Vault modelling, and the star schema, organisations can address various needs, whether it's reducing redundancy, integrating multiple data sources while preserving history, or optimising query performance.

Key Takeaways

- **Takeaway 1:** Data modelling is essential for organising data logically, improving efficiency, and supporting real-time analytics.
- **Takeaway 2:** Third Normal Form reduces redundancy, improves integrity, and is best for operational systems requiring high accuracy.
- **Takeaway 3:** Data Vault modelling enables scalability and historical tracking when integrating multiple and changing data sources.
- **Takeaway 4:** Star schema simplifies complex queries, optimises performance, and is ideal for analytical environments.